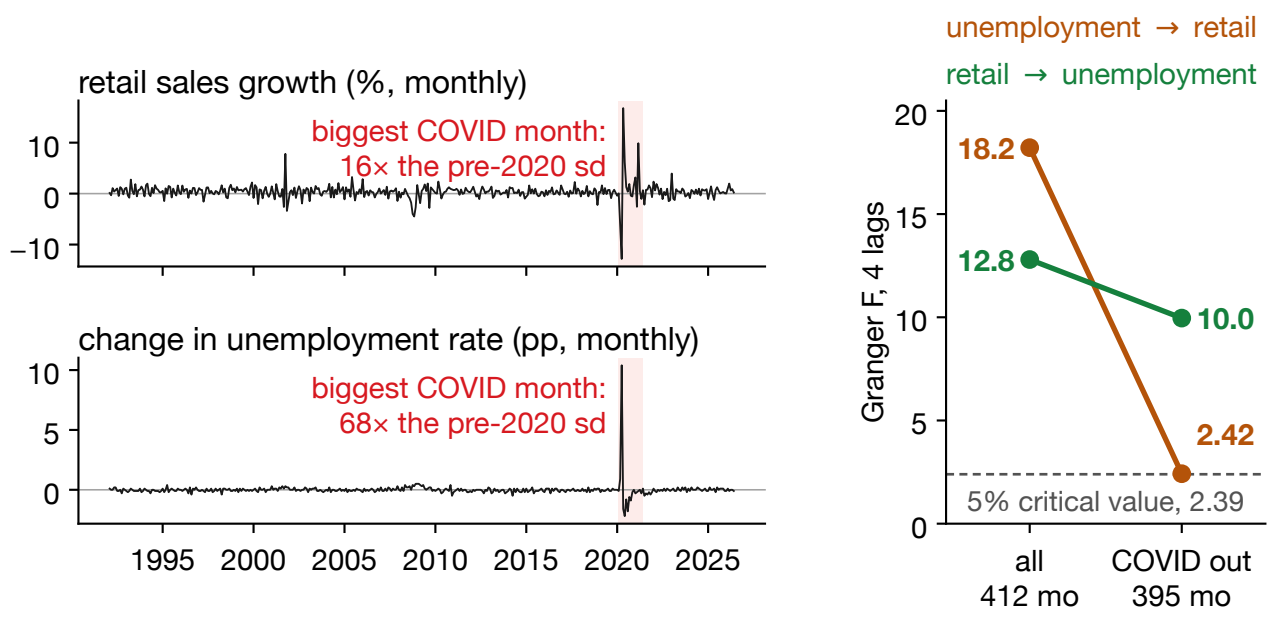
retail sales growth (%, monthly)
biggest COVID month:
16× the pre-2020 sd
10
0
−10
change in unemployment rate (pp, monthly)
biggest COVID month:
68× the pre-2020 sd
10
5
0
1995
2000
2005
2010
2015
2020
2025
unemployment → retail
retail → unemployment
Granger F, 4 lags
20
15
10
5
0
18.2
12.8
10.0
2.42
5% critical value, 2.39
all
412 mo
COVID out
395 mo